BİL462 Biometric Recognition · Project Phase 3 · Data & Preprocessing Checkpoint · Topic A5

Password-Typing Verification Across Sessions on the CMU Keystroke Benchmark

Bahadır İhsan Herdem (221101034) · Ömer Efe Öztürk (221101059)

Covered today: ① dataset statistics · ② sample preprocessing outputs · ③ observed quality issues · ④ samples rejected during preprocessing
Data in hand: **DSL-StrongPasswordData.csv** from cs.cmu.edu/~keystroke (Killourhy & Maxion, DSN 2009). Everything shown is reproduced by `proje/checkpoint.py`.

① Dataset statistics

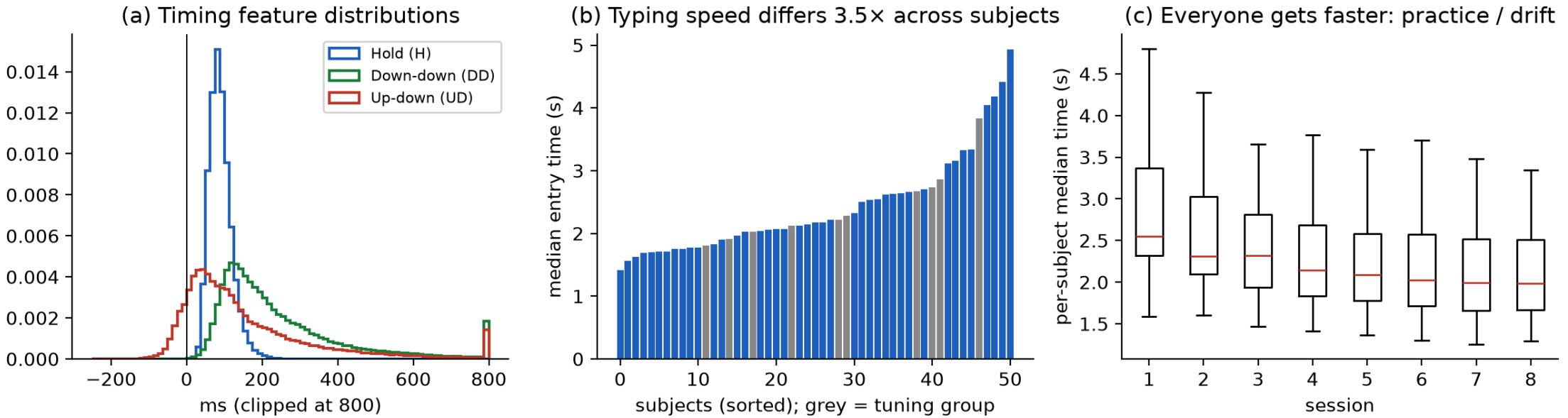
51
subjects (s002-s057, 5 IDs absent)

8 × 50
sessions × repetitions, ≥ 1 day apart

20,400
password entries, 0 missing

31
features: 11 hold (H), 10 down-down (DD), 10 up-down (UD)

2.30 s
median time to type **.tie5Roanl**+Enter



Mean ± SD: H 90 ± 30 ms, DD 249 ± 218 ms, UD 159 ± 222 ms · per-subject median entry time 1.41–4.93 s · session median 2.80 s (S1) → 2.08 s (S8); 49/51 subjects faster in S8 · split: 10 tuning subjects (grey) / 41 evaluation subjects.

Preprocessing pipeline and validation checks

Load CSV
subject, session, rep + 31 timings (s)

→

Validate
structure, completeness, physical consistency (table)

→

Index
array [subject][session 1–8][rep 1–50][31]

→

Enrolment-only rejection
drop atypically slow training entries (slide ④)

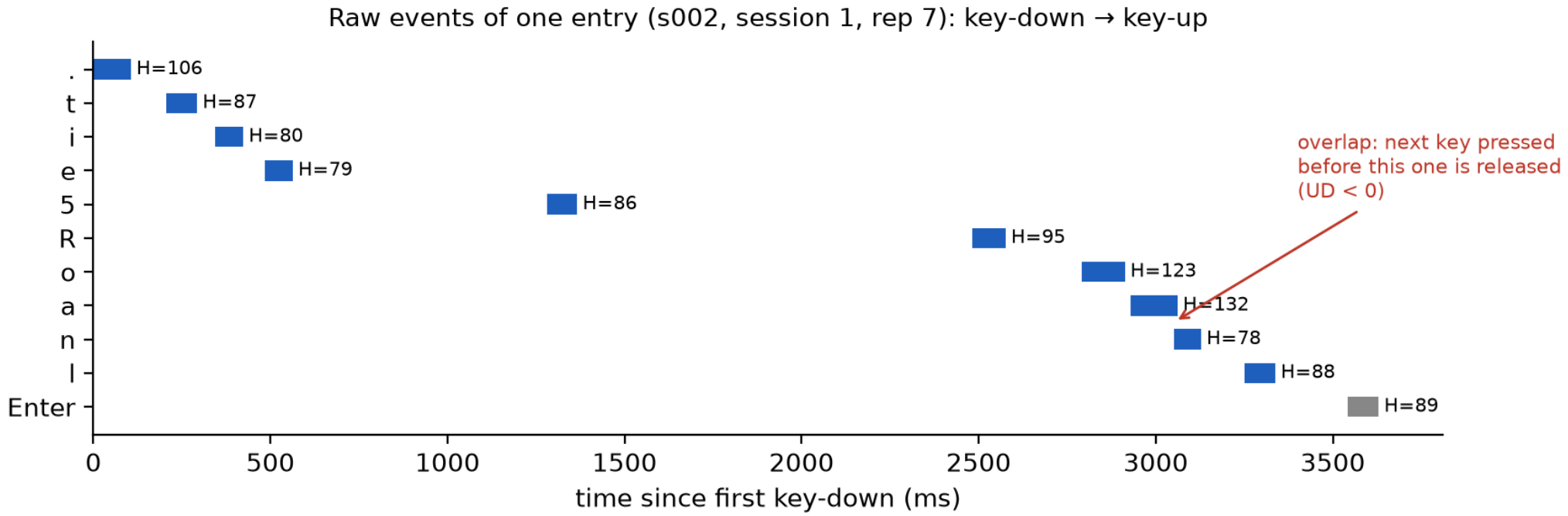
→

Per-user normalisation
scaled Manhattan: ÷ MAD · SVM: z-score · fitted on enrolment only

Check	Result	Detail
columns: 3 index + 31 timing	**pass**	11 H, 10 DD, 10 UD
missing / non-numeric values	**pass**	0
duplicate (subject, session, rep)	**pass**	0
complete grid 51 x 8 x 50	**pass**	51 x 8 x 50
hold times > 0	**pass**	min 1.4 ms
down-down latencies > 0	**pass**	min 1.1 ms
identity DD = H + UD	**pass**	max deviation 8.9e-16 s
negative up-down (key overlap)	**pass**	22118 values (10.8%), valid, kept

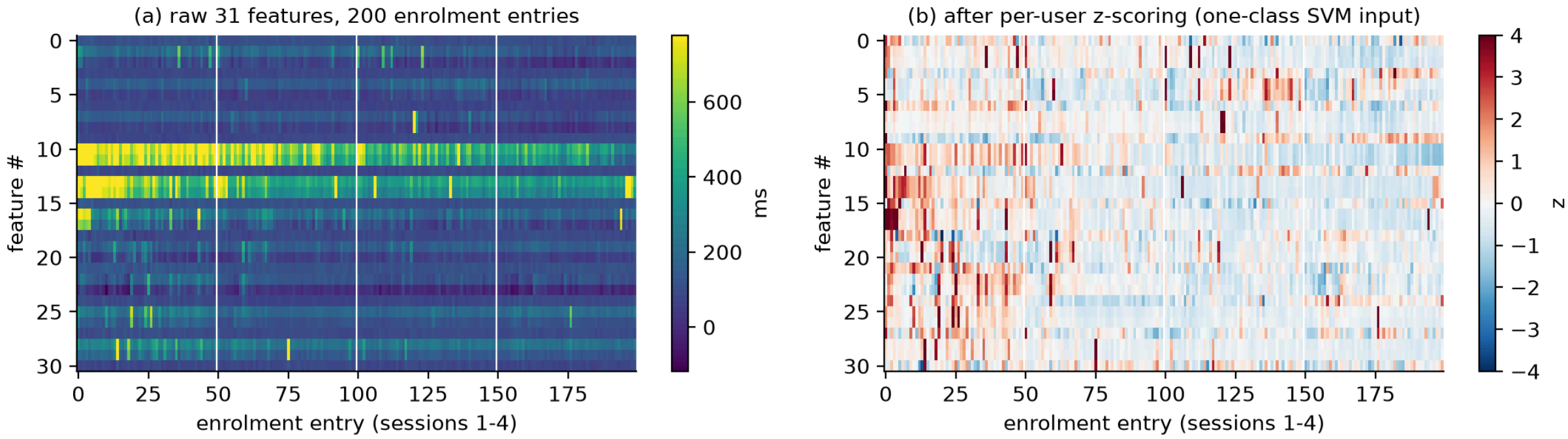
- No keystroke detection step exists: the CMU recorder accepted an entry only if the 10 characters were typed **correctly**; entries with typos were retyped and not stored.
- Timestamps from an external reference clock (± 200 µs, paper §4.2); values stored with 0.1 ms resolution.
- Test entries are **never** filtered or used to fit normalisation.

② Sample preprocessing output: one entry → one 31-D vector



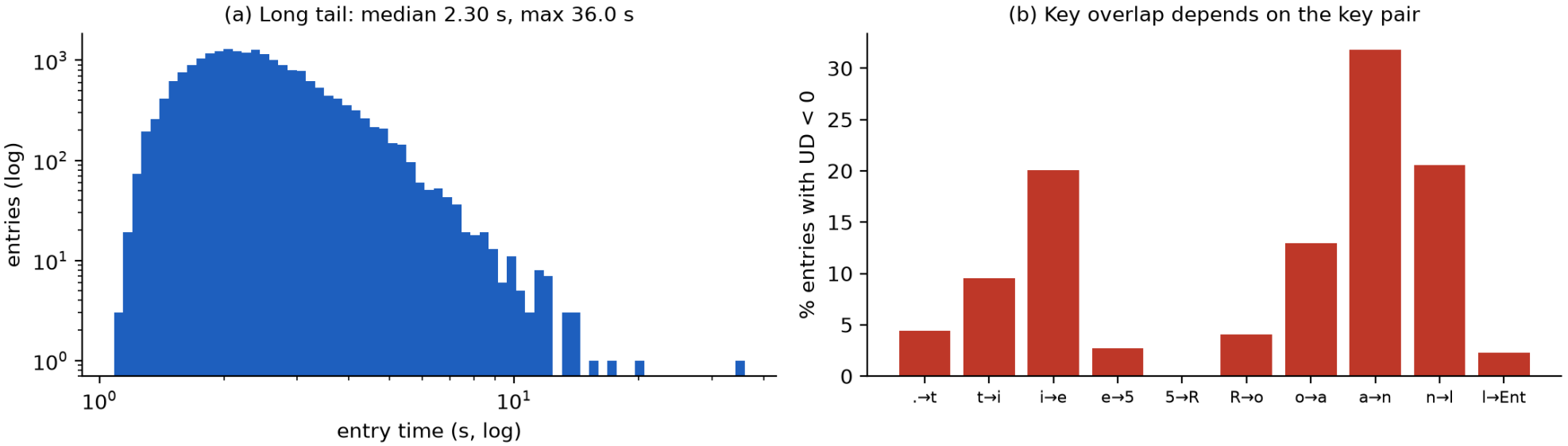
From key-down / key-up events: **H** = up – down of the same key, **DD** = down(next) – down(this), **UD** = down(next) – up(this) = DD – H. Shift+R counts as one keystroke (R), Enter is the 11th key. The vector is stored in key order: H.period, DD.period.t, UD.period.t, H.t, … , H.Return.

② Sample preprocessing output: one user's enrolment template



s002, condition C2 (cross-session): 200 enrolment entries = sessions 1–4 (white lines). Raw features differ by an order of magnitude (hold ≈ 90 ms, pauses before **5** and **R** ≥ 600 ms), so raw Manhattan distance is dominated by a few long latencies. After per-user z-scoring every feature has equal weight; the early entries (left) are visibly slower, the drift we measure later.

③ Observed quality issues



- **Key overlap:** 10.8% of UD values are negative (54% of entries have ≥1); per subject 0%–35%. Real typing behaviour, **kept**.
- **Long pauses:** 189 entries contain a latency > 2 s; longest entry 36.0 s (s049, S1, one pause of 26.0 s).
- **Practice effect / drift:** median entry time falls 2.80 → 2.08 s; within-session variation is also large (mean CV 0.18).
- **Redundant features:** DD = H + UD exactly, so a user's 31×31 covariance has rank 21 → Mahalanobis needs a ridge term.
- **Lab conditions:** one laptop + external keyboard, 50 entries in a row; browser timing in our demo will differ, so demo accuracy is not evidence.
- **Ground truth limits:** subject IDs have gaps (5 absent IDs, as released); typo behaviour is not recorded.

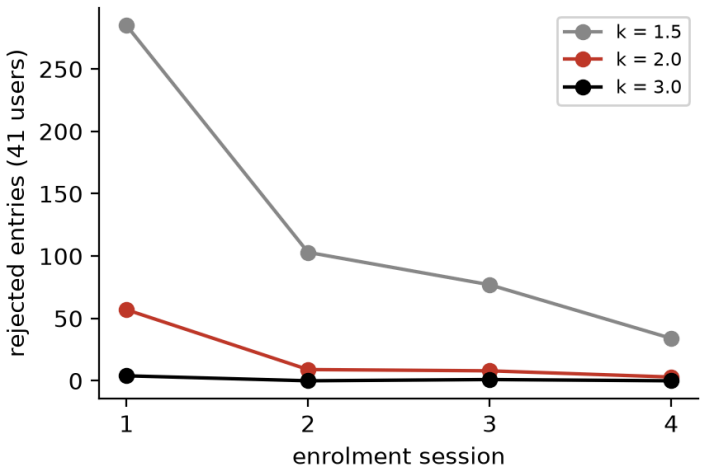
④ Samples rejected during preprocessing

Rule (enrolment entries only): reject an entry whose total time exceeds **k × the user's own median** over that user's enrolment entries. Chosen value **k = 2**; computed without any test data.

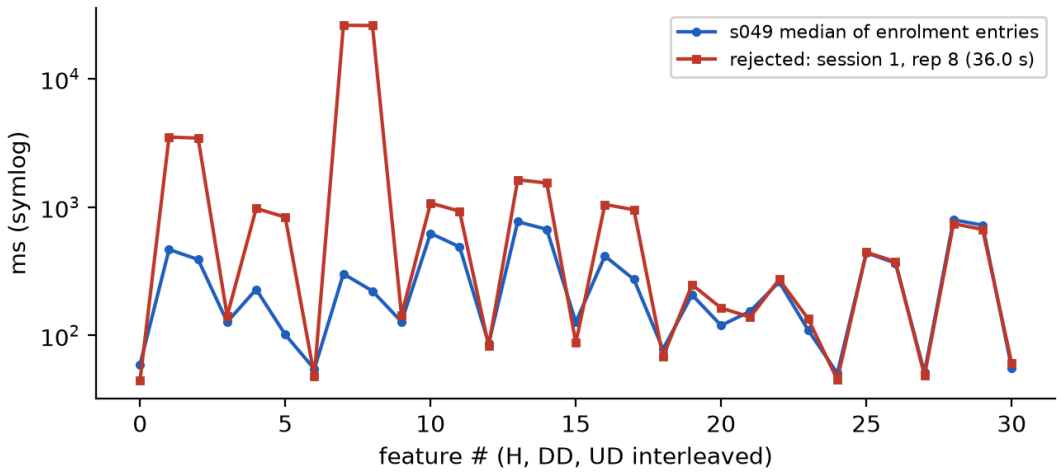
k	C2 rejected	users affected	max / user	C1 rejected
1.5	499 / 8200 (6.1%)	41 / 41	44	332 / 8200
2.0	77 / 8200 (0.9%)	25 / 41	15	38 / 8200
3.0	5 / 8200 (0.1%)	4 / 41	2	3 / 8200

C2 = 41 users × 200 enrolment entries (S1–S4); C1 = 41 × 8 sessions × 25 entries.
Rejections fall mostly on session 1 (first contact with the password).
Largest rejected entry: s049 S1 rep 8: 36.0 s vs. user median 5.5 s (DD.i.e = 26.0 s).

(a) Rejections concentrate in session 1



(b) A rejected entry: long pauses inflate DD/UD, holds unchanged



④ Does rejection matter? First check on C2 (cross-session)

Detector (41 eval. users)	mean EER, all enrolment	mean EER, k = 2 rejection	difference [95% CI, paired subject bootstrap]
Manhattan	22.3%	22.0%	-0.24 pp [-0.41, -0.08]
Scaled Manhattan	15.1%	15.0%	-0.08 pp [-0.20, +0.03]
Mahalanobis	15.3%	15.1%	-0.22 pp [-0.48, +0.00]
One-class SVM	14.3%	14.1%	-0.18 pp [-0.47, +0.04]

- Removing ≈ 1% of enrolment entries lowers EER by only 0.1–0.25 percentage points; the CI includes 0 for three of four detectors.
- The error is dominated by **drift between sessions**, not by a few bad entries; cleaning cannot fix that.
- **Decision:** headline results use all entries (same as the reference study, needed for the C3 comparison with Killourhy & Maxion); k = 2 rejection is reported as a separate, labelled variant.

Status and next steps

- **Done:** data in hand and validated; preprocessing + rejection module (`keystroke/preprocess.py`); 3 distance detectors + one-class SVM ($\nu = 0.05$, $\gamma = 0.03$ from 10 tuning subjects); reference replication matches the paper (scaled Manhattan 9.6%, Manhattan 15.3%).
- **By Nov 27 (milestone):** score distributions and DET curves for within-session (C1) vs. cross-session (C2), per-user EER, bootstrap CIs; drift curve S1 → S8.
- **Then:** report score fusion over k = 3, 5, 10 consecutive entries (already implemented); optional Siamese embedding trained on subjects disjoint from evaluation; localhost login demo (team data only after instructor approval, never mixed into evaluation).
- **Open question for the instructor:** approval to collect the two team members' own typing data for the demo.

AI use: Claude Code (Anthropic) assisted with analysis code, figures and slide layout; all numbers were regenerated from our scripts and checked against the CSV and the reference paper.